Andrew McKnight

resume@andrew.mcknight.io · https://mcknight.io

Experience

➤ **Sentry** · Senior Software Engineer · Remote **Jan 2021 – Current**

- **Stack**: Objective-C++, Swift, Python, SwiftUI, Django, Claude Code, SPM, XcodeGen, CocoaPods, Carthage, Swiftlint, clang-format, CodeCov, pre-commit, Fastlane, Arkana, GitHub Actions, Redash, SauceLabs, TestFlight, Linear
- **Deliverables**
  - Sentry Cocoa SDK: mobile app crash and performance monitoring
    - Integrated Specto's profiler into Sentry's Cocoa SDK and iterated with new API/features
    - Built User Feedback widget, the first user-facing UI shipped in Sentry Cocoa SDK
    - Built demo apps for SDK and Sales teams flexible enough to simulate any customer configuration
    - Parallelized long-running CI jobs and modularized test suites to find and fix statefulness issues
  - Seer for iOS and macOS: proposed and built native apps to manage our agentic offerings
  - Billing: platformized legacy system for product teams to self-serve how they charge for features

➤ **Specto** · Senior Software Engineer · Remote **Nov 2019 – Jan 2021**

- **Stack**: Objective-C++, Swift, Go, Python, Protobuf, Jinja, XcodeGen, Bazel, pre-commit, GitHub Actions, Google Cloud Run
- **Highlights**
  - First employee at startup and remained through acquisition by Sentry
- **Deliverable**: Specto SDK, mobile app profiling SDK for iOS and Android
  - Shared C++ core between iOS and Android with Protobuf RPC to Python/Go backend
  - Python scripts to generate scoped Xcode projects from Bazel targets (pre `rules-xcode`)
  - Hardened SDK init sequence, app launch profiling, NSURLProtocol hooks, MetricKit ingestion
  - Signal and mach exception propagation with open source contributions to other crash reporters
  - Smoke test generation validating integration across CocoaPods, Carthage, Bazel and SPM

➤ **Twitter** · Software Engineer · Cambridge, MA **Apr 2015 – Dec 2016**

- **Stack**: Objective-C, C, Swift, Java, Scala, Clojure, Coffeescript, Rails, React, Protobuf, Storm, Heron, Cascading, Scalding, Graphite, Beanstalk, Kafka, Redis, Cassandra, AWS, TeamCity, Chef, Trello, Vertica, Zeppelin
- **Highlights**
  - Presented NSOperation rearchitectures and open source tools at MobileDevCon 2016 in Amsterdam
  - Work on high-volume lambda backend, including on-call and migration from Storm to Heron
- **Deliverables**
  - Crashlytics SDK: crash reporting and beta deployment SDK for Apple platforms and Android
    - Rearchitected CLIs to bootstrap projects, upload dSYMs and interop with our macOS app
    - Added Bitcode and tvOS compatibility to SDKs, and investigated watchOS support
    - Researched and built feature surfacing solutions to detectable ecosystem-wide crash causes
  - Fabric SDK and macOS app: developer tool ecosystem management
    - Streamlined build and release processes with TeamCity agents deployed via Chef
    - Rearchitected macOS app launch sequence using queues of async/compound NSOperations
  - FABConfig: standardizes config of all Xcode projects and released the open source FABConfig
  - FABOperation: custom NSOperation subclasses for async work and grouping tasks

Presentations

➤ **iOSDevCampCO** · Colorado Springs, CO **May 2018**
Colloquial Swift (45 min)

- Presented findings from analysis of common function implementations in open source repositories as a way to generate possible Swift Evolution proposals

➤ **CocoaHeads Boston** · MIT, Cambridge, MA **Jun 2017**
UIViewController Decomposition and UI Reuse (20 min)

- Discussing strategies to separate responsibilities from UIViewControllers and a library of reusable UI components, e.g. animated transparent modal presentation with a close button

➤ **mDevCon** · Amsterdam, NL **Mar 2016**
Command and NSOperation (45 min)

- Delivered presentation on the Command software design pattern, and NSOperation
- Introduced subclasses we open sourced at Crashlytics for asynchronous and compound operations
- Previously delivered at a lunchtime tech talk at Twitter, Cambridge, MA (Nov 2015) and CocoaHeads Boston meetup, MIT, Cambridge, MA (Dec 2015)

Mentorship

➤ **Girls Who Code** · Twitter, Cambridge, MA **Jul 2016**
iOS Dev Session

- Part of GWC summer internship covering programming first principles to full-stack style development
- Designed and led iOS session where we built a Flappy Bird clone, introducing them to Xcode as an IDE, Interface Builder, Playgrounds and Swift